

AKHIL SANTHA

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PROFESSIONAL SUMMARY

Senior Machine Learning Engineer with 5+ years of experience delivering production-grade ML and GenAI systems. Led development of scalable MLOps pipelines and LLM-driven solutions across AWS and Azure, improving model performance and reducing deployment latency. Strong track record in translating business problems into high-impact ML systems across NLP and computer vision domains.

PROFESSIONAL EXPERIENCE

CVS, TX | *Senior Machine Learning Engineer*

Jun 2025 – Present

- Developing machine learning models for SKU assortment, leveraging historical sales and transactional data to optimize inventory and drive revenue growth.
- Developing deep learning models and applying GenAI / prompt engineering to generate product embeddings and structured product attributes from unstructured descriptions, supporting advanced analytics and recommendation systems aligned with business objectives.
- Designing and implementing MLOps pipelines with Kubernetes to deploy models into production, ensuring scalability, reliability, and monitoring of ML workflows.
- Building and maintaining efficient data pipelines to support ML operations, improving data accessibility and reducing processing time for downstream models.
- Developing and leading a team to build optimized, scalable, and maintainable ML pipelines, enforcing best practices.

FANNIE MAE, TX | *Machine Learning Engineer II*

Jun 2023 – Jun 2025

- Led the development and deployment of deep learning models for financial data analysis, enabling anomaly detection, pattern recognition, and operational improvements.
- Built and optimized large language models using AWS Bedrock and implemented advanced algorithms (Isolation Forests, Random Cut Forest, LSTM Autoencoders, Prophet) to generate actionable insights and support predictive analytics.
- Designed and implemented end-to-end MLOps pipelines using AWS SageMaker for model training, deployment, monitoring, and seamless integration with production systems.
- Developed high-performance prediction APIs using Flask, improving invocation speed through parallel computing and ensuring smooth integration with user-facing applications for enhanced decision-making.
- Directed data engineering initiatives with AWS Glue and TensorFlow to streamline pipelines, reduce processing time by 20%, and deploy scalable, production-ready ML solutions for financial analytics.

SMITH MICRO SOFTWARE, CA | *Machine Learning Engineer*

Jun 2021 – Apr 2023

- Led the design, training, and deployment of sequence-to-sequence deep learning models (LSTMs, Transformers) for large-scale parental control systems, achieving 99% accuracy in bullying and profanity classification and significantly improving user safety.
- Developed and integrated a BERT-based MAC address classification system into iOS applications, enhancing device identification accuracy and streamlining device-level controls.
- Optimized and quantized deep learning models to 8-bit precision, reducing model size by 75% while retaining over 95% of baseline accuracy, allowing efficient deployment on resource-constrained edge devices. Worked on deploying the trained models onto iOS devices using COREML tools.
- Led the Location Anomaly Detection initiative, applying supervised and unsupervised learning techniques to identify abnormal behavioral patterns, improving user safety signals and system reliability.
- Designed and fine-tuned action recognition models optimized for edge environments, balancing accuracy, latency, and memory constraints.
- Deployed and maintained cross-platform ML services using AWS and TensorFlow, supporting scalable inference and continuous model updates.

- Developed deep learning classification models on DICOM-formatted IVUS images to accurately identify and classify stent regions, supporting catheter-based diagnostic workflows.
- Designed and implemented segmentation models for IVUS images to precisely detect stent and calcium regions, improving the accuracy of intravascular image analysis. Applied image processing and computer vision techniques to improve feature extraction and model performance in medical imaging pipelines.
- Conducted research and experimentation with 2D and 3D U-Net architectures to enhance stent segmentation performance across varying imaging conditions and modalities.
- Implemented distributed training pipelines using PyTorch, enabling scalable multi-GPU training and reducing model training time for large IVUS datasets. Optimized trained models through techniques such as quantization to enable efficient deployment on medical devices while maintaining clinical-grade accuracy.

- Developed and trained deep learning models using TensorFlow for face and eyeglass detection, achieving 98% accuracy in eyeglass classification. Applied transfer learning techniques by fine-tuning the Inception ResNet v1 architecture, improving model accuracy and inference efficiency for real-world deployment.
- Optimized and quantized trained models using TensorRT with 8-bit precision, reducing model size by 75% while maintaining equivalent detection performance.
- Designed and deployed inference pipelines using OpenCV and DLib to integrate face, eyeglass, and landmark detection models into kiosk-based systems, delivering interactive and personalized user experiences.
- Developed models for classification and segmentation of 3D point cloud data, implementing point cloud registration and convex hull construction using PCL and Open3D.

- Designed and developed deep learning models for skin lesion classification, outperforming state-of-the-art benchmarks by 2% on publicly available datasets.
- Built and optimized TensorFlow-based computer vision models robust to variations in skin tone, lighting conditions, and lesion appearance, ensuring consistent performance across diverse populations.
- Applied advanced image processing techniques, including edge detection and morphological operations, to accurately extract lesion regions and improve downstream classification accuracy.
- Developed and fine-tuned algorithms to normalize color and texture variations across images, enhancing model generalization and reliability.

SCHOLARLY WORKS & PUBLICATIONS

- **Publication:** [Deep Learning Methods for Sign Language Translation](#). This article delves into different deep learning architectures and models used for sign language translation.
- **Thesis:** [Deepfakes Generation using GANs](#). This research focuses on using deep neural networks to generate manipulated faces in images and videos.

EDUCATION

Master of Science in Computer Engineering
Rochester Institute of Technology, NY

Jan 2018 – May 2020

Bachelor of Science in Electronics and Communications Engineering
University College of Engineering, Osmania University, India

Sep 2013 – May 2017

SKILLS

- **Machine Learning:** Representation Learning, Transfer Learning, Computer Vision, Deep Learning, NLP
- **ML Systems:** Distributed Training, LLM Training & Inference Infrastructure, Model Parallelism, Checkpointing, Data Parallelism, GPU Training
- **Cloud & Infra:** AWS, GCP, Azure, Docker, Kubernetes, CI/CD, MLOPs, Streamlit, TensorBoard
- **Programming & Packages:** Python, C++, Swift, NumPy, Pandas, Scikit-Learn, OpenCV, Linux, Git
- **Frameworks:** PyTorch, JAX, TensorFlow, Ray, TensorFlow Lite, PyTorch, Hugging Face, TensorRT, Keras